

DECODED EXPLAINED

19 July, 2026

Knowledge economy vs Agentic Economy

Knowledge Economy: A world where being valuable means knowing a lot about one subject deeply.

Agentic AI Economy: A world where being valuable means judging and checking AI's work, since AI can now plan and finish tasks on its own.

<u>Point of Comparison</u>	<u>Knowledge Economy</u>	<u>Agentic AI Economy</u>
What made a person valuable	Knowing a lot about one subject, like a doctor knowing medicine deeply	Good judgement ,knowing what's right, wrong, or missing, even outside your own subject
Main daily work	Doing the task yourself, from start to finish	Checking and guiding AI's work, since AI does much of the task itself
Biggest danger	Not knowing enough , gaps in your own knowledge	Trusting a wrong AI answer too easily, just because it sounds confident
Type of skill needed	Deep expert knowledge in one narrow area	Broad understanding across many areas, even ones you're not an expert in
Role of computers/AI	Helped store information and gave it to you fast, but you still had to think and decide	Can finish tasks almost on its own, without constant checking

With the present state of development, Artificial Intelligence can effectively do which of the following?(2020)

1. Bring down electricity consumption in industrial units
2. Create meaningful short stories and songs
3. Disease diagnosis
4. Text-to-Speech Conversion
5. Wireless transmission of electrical energy

Select the correct answer using the code given below:

- (a) 1, 2, 3 and 5 only
- (b) 1, 3 and 4 only
- (c) 2, 4 and 5 only
- (d) 1, 2, 3, 4 and 5

Context

India achieved a historic milestone as Skyroot Aerospace's Vikram-1 became the country's first privately developed orbital launch vehicle to successfully place satellites into orbit.

Background

- India's space sector was opened to private participation in 2020.
- Skyroot Aerospace became India's first private company to achieve orbital launch capability through Vikram-1.
- The mission successfully deployed satellites into orbit, validating India's private launch capability.

IN-SPACE – Government body that gives permission and support to private companies wanting to work in space; started in 2020.

NSIL- ISRO's business arm; markets rockets and satellites, and handles new commercial deals with private partners; example - Skyroot Vikram 1

Antrix– ISRO's oldest marketing company; started in 1992; now mainly handles older contracts, and major missions with other countries, example - PSLV C37 (carrying 102 satellites)

Vikram-S

- India's first privately built rocket, launched by Skyroot Aerospace.
- Launched on 18 November 2022.
- **It was a suborbital rocket , it touched space but didn't go around Earth.**
- It was small , just one stage, about 6 metres tall.
- **It carried no satellites** , it was only a test flight.
- Its importance: proved that a private Indian company could build and launch a working rocket.

Vikram-1

- India's first privately built orbital rocket, also by Skyroot Aerospace.
- Launched on 18 July 2026.
- **It is an orbital rocket , it actually went around Earth.**
- It is bigger , four stages, about 20 metres tall.
- **It carried real satellites and placed them into orbit.**
- Its importance: made India the 3rd country in the world (after the USA and China) to have a private company launch something into orbit.

Why Vikram-1 is a Historic Achievement

- India's First Private Orbital Launch
- Shows that private firms can complement ISRO.
- Boosts the Space Economy - Encourages domestic and foreign investment.
- Reduces dependence on foreign launch providers.

Context

A study in Assam's Raimona National Park revealed that invasive plant species like *Lantana camara* and *Chromolaena odorata* act as vital nectar and larval host sources for butterflies. Despite harming native vegetation, these plants sustain butterfly populations by blooming for prolonged periods when native wildflowers are scarce.

What is an Invasive species

An invasive alien species is a plant or animal that doesn't naturally belong to a place, but gets brought there ,and once it arrives, it spreads fast and starts harming the local plants, animals, or environment.

Eg - Water Hyacinth in Kerala ,
It came from South America



Larval Host Plants - Larval host plants are the specific plants where adult butterflies lay their eggs, and which caterpillars eat after hatching.,

Nectar Plants - Nectar plants are flowering plants that provide sugar-rich nectar for adult butterflies. Purpose is to get Instant energy for flight, mating, and migration.

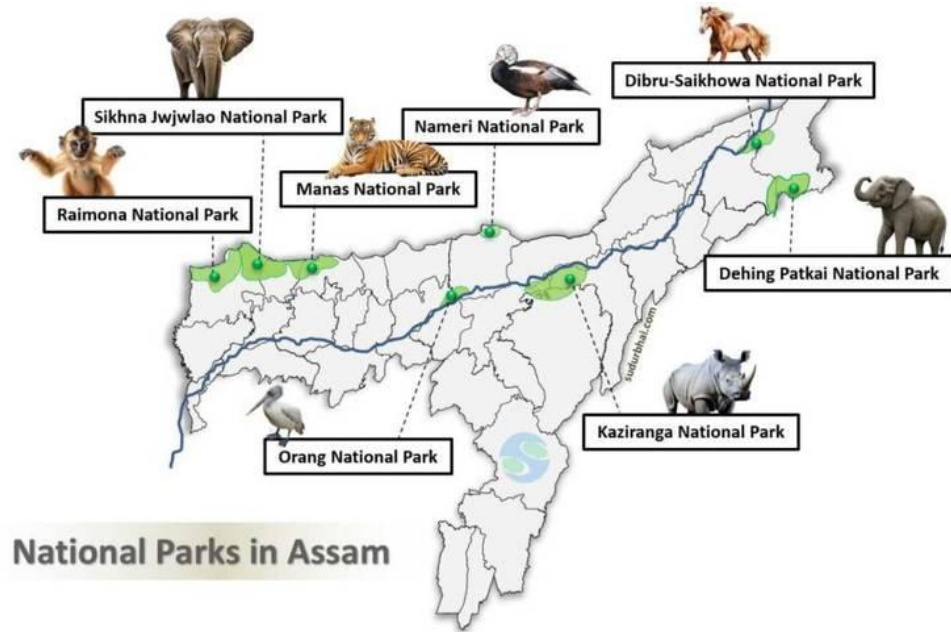
- **Nectar and Larval Hosts:** *Lantana camara* attracted over 30 butterfly species , while *Chromolaena odorata* supported 24 species. Both serve as vital sources of nutrition for adult butterflies and their larvae.
- These are invasive species (from North, South and Central America)
- **Conservation Strategy Balance:** *The findings emphasize moving away from blanket eradication toward carefully managed patches of these invasives to maintain continuous floral resources in protected areas.*

Raimona National Park

- **Location:** Raimona National Park is in **Kokrajhar** district, **Assam**, right next to the **India-Bhutan border**.
- When it was made: It became Assam's 6th national park in June 2021.
- Raimona National Park, is home to the indigenous **Bodo tribe**
- Why it's special: It protects the endangered golden langur, plus elephants, tigers, and over 150 species of butterflies ,and it **connects with forests in Bhutan to form one big cross-border wildlife area**.

Assam now has 8 national parks in total.

- Kaziranga National Park
- Manas National Park
- Nameri National Park
- Orang National Park
- Dibru-Saikhowa National Park
- Dehing Patkai National Park
- Sikhna Jwhlwao National Park



2018 pyq

Why is a plant called *Prosopis juliflora* often mentioned in news?

- (a) Its extract is widely used in cosmetics
- (b) It tends to reduce the biodiversity in the area in which it grows
- (c) Its extract is used in the synthesis of pesticides
- (d) None of the above

Context

The Non-Animal Testing Models (NAMs) are emerging as scientifically credible alternatives and complements to the traditional animal testing, presenting an opportunity for India to further strengthen its pharmaceutical innovation ecosystem.

Some animals that is used for drug testing

Caenorhabditis elegans (C. elegans) – A tiny worm, about 1mm long. - Its whole body is see-through, so scientists can watch its cells work in real time. It has a short life ,and use to study aging, genes,

Drosophila melanogaster – The common fruit fly. Breeds very fast (a new generation every 10 days).60% of its genes are similar to human genes linked to diseases

Zebrafish – A small, striped freshwater fish-Its babies (embryos) are see-through, so scientists can watch organs form. It shares about 70% of its genes with humans.

What are Non-Animal Testing Models (NAMs)?

Non-animal testing methods (NAMs) are scientific approaches that use human-relevant biological systems and computational methods to generate safety, efficacy and toxicity information, instead of relying solely on animal testing.

What are stemcells?

- They are special cells that can grow into many different types of body cells (like liver cells, brain cells, or heart cells).
- Scientists grow them into organoids (organoid is a tiny, mini version of a real body organ). Using stem cells, labs can grow tiny, mini versions of real organs , like a mini-liver or mini-brain , inside a dish.
- Since they behave like small working organs, they can be used to see how a new medicine affects the body.
- This removes the need for animal testing. Instead of testing new drugs on animals, scientists test them on these lab-grown organoids , this is the core idea behind Non-Animal Testing Models (NAMs).

Types of NAM platforms

- **Organoids:** Miniature three-dimensional structures grown from human stem cells that mimic the architecture and function of human organs.
- **Organ-on-chip systems:** Microfluidic devices that recreate the physiological environment of human tissues and organ interactions.
- **Three-dimensional human cell culture models:** Advanced cellular systems that better replicate human tissue behaviour than conventional two-dimensional cultures.
- **Computational toxicology:** Computer-based models that predict chemical toxicity and drug responses using biological and chemical data.
- **Artificial Intelligence (AI)-enabled models:** Machine learning systems that analyse large biomedical datasets to predict human drug responses.

Why is a Shift Needed?

- **Limitations of conventional animal testing:** Animal physiology, metabolism, immune responses and disease mechanisms differ from those of humans
- **Ethical Considerations:** These models reduce or eliminate the need for live animals (such as rodents and primates) in painful or fatal testing procedures.
- **India's Existing Strengths:** India possesses a robust generic pharmaceutical industry, enabling it to emerge as a major global supplier of affordable medicines.

Generic drug – A copy of a normal chemical medicine after its patent ends. Since it's a simple chemical, it can be copied exactly, just like photocopying a document.

Biosimilar – A copy of a biologic medicine (a medicine made from living cells, not chemicals). Since living things are complex, you can't copy them exactly , a biosimilar is only "highly similar," not identical, to the original.

Biosimilar example

Example: Insulin for diabetes

The original biologic medicine is called **Humalog**, it's made using living cells (genetically modified bacteria or yeast) that are trained to produce insulin, just like how a human body naturally makes insulin.

When Humalog's patent ended, other companies wanted to make a **cheaper version**. But since insulin comes from living cells (not simple chemicals), they couldn't copy it exactly, no two batches of living cells behave in 100% identical ways.

So they made a biosimilar called **Admelog**. It works almost the same way as Humalog, lowers blood sugar in a similar way, and is much cheaper, but it isn't a perfect identical copy.

Because it's not identical, biosimilars need extra testing (clinical trials) before they can be sold, to prove they are safe and work almost the same as the original.

THANK YOU